

Computer Organization and Architecture		
Lab #:	6	
Lab Title:	Enter two integers: n1 and n2 and prints the sum of all numbers	
Submitted by:		
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Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

LAB: 06

Task 1:

Write a program that asks the user to enter two integers: n1 and n2 and prints the sum of all numbers from n1 to n2. For example, if the user enters n1=3 and n2=7, then the program should display the sum as 25.

Code:

```
.data
prompt1: .asciiz "Enter the 1st integer n1: "
prompt2: .asciiz "Enter the 2nd integer n2: "
result_msg: .asciiz "The sum from n1 to n2 is: "

.text
.globl main
main:
    li $v0, 4      # syscall for print string
    la $a0, prompt1  # load address of prompt1
    syscall

    li $v0, 5      # syscall for read integer
    syscall

    move $t0, $v0   # store n1 in $t0

    li $v0, 4
    la $a0, prompt2  # load address of prompt2
    syscall

    li $v0, 5
    syscall

    move $t1, $v0   # store n2 in $t1
```

```

# Initialize sum to 0
move $t2, $zero    # sum = 0
move $t3, $t0      # i = n1

sum_loop:
    bgt $t3, $t1, end_sum_loop # if i > n2, exit loop
    add $t2, $t2, $t3          # sum = sum + i
    addi $t3, $t3, 1           # i = i + 1
    j sum_loop                 # repeat loop

end_sum_loop:
    li $v0, 4
    la $a0, result_msg
    syscall

    # Print the sum
    li $v0, 1
    move $a0, $t2              # move sum to $a0
    syscall
    li $v0, 10                 # syscall for exit
    syscall

```

Output:

Mars Messages	Run I/O
	Enter the 1st integer n1: 14 Enter the 2nd integer n2: 29 The sum from n1 to n2 is: 344 -- program is finished running -- Reset: reset completed. Enter the 1st integer n1: 3 Enter the 2nd integer n2: 6 The sum from n1 to n2 is: 18 -- program is finished running --